



Free radical mechanism for reactive processing of polymers in the presence of organoclay/peroxide nanoreactors: Grafting maleic anhydride onto polypropylene

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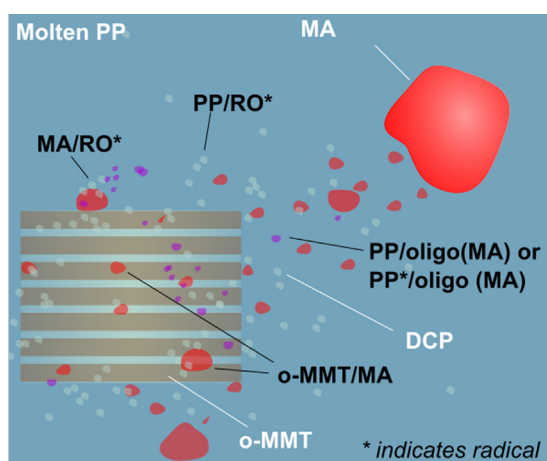
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HIGHLIGHTS

- Real nanoreactor must have o-MMT/DCP mass ratio >4:1.
- Succinic groups with different lengths are grafted onto PP-graft-MA.
- Succinic groups are grafted preferentially to PP end chains.
- The presence of o-MMT induces secondary PP macroradicals recombination.
- Presence of o-MMT induces MA homopolymerization.

GRAPHICAL ABSTRACT



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ABSTRACT

PP-graft-MA is a common PP compatibilizer, produced via reactive extrusion using peroxides as initiator, which causes severe molar mass reduction. To control PP degradation and increase maleic anhydride grafting, Shi and coworkers (2006), proposed the use of nanoreactors. In this paper we propose a complete new reactive mechanism involving nanoreactors during PP-graft-MA reactive processing. The samples were prepared via melt mixing using nanoreactors with different organoclay: peroxide ratios (oMMT:DCP). Organic MMT showed to absorb not more than 16 wt% of DCP, i.e., diffusional controlled release of alkoxy radical will not be observed at o-MMT:DCP ratios < 5:1. The presence of o-MMT tend to increase MA homopolymerization, which consumes the initiator, interfering with the grafting of succinic groups onto the PP backbone, due to increased hindrance of oligo (MA) formed, thereby reducing degradation of the PP chain. Thus, the use of nanoreactors promotes MA homopolymerization rather than grafting as previously proposed in the literature.

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1. Introduction

Polypropylene (PP) is one of the most used polymers in the world, due to its low cost and density. However, PP lacks some properties required for certain applications, and frequently needs to be compounded with other materials. When the second phase added to PP is polar interfacial bonding will be poor and premature failure is expected. Grafting polar groups onto PP is an alternative to improve interfacial bonding and reduce, or eliminate, failure caused by poor compatibility of the phases.

In the 1960s, the use of bifunctional monomers to modify polyolefins was proposed (Chang and White, 2003). Some rules were established at that time, among these, that the monomer should have low or no tendency for undergoing free radical polymerization (Chang and White, 2003). Hence, other reactions induced by free radical reactions occur more readily than the reactions involving the monomers. In fact, polymer chain degradation is observed to a high extent. Changes in monomer reactivity are possible by adding to the medium molecules that may act as charge transfer medium, such as styrene (ST). This system has been studied by Hu and coworkers (Hu et al., 1993) and by Bettini & Ruvolo-Filho (Bettini and Ruvolo Filho, 2008); both investigated the grafting of maleic anhydride (MA) onto PP in the presence of ST. Bettini & Ruvolo-Filho reported the formation of polypropylene grafted with a copolymer ST-MA and lower molar mass reduction, as described by Hu and coworkers (Hu et al., 1993).

Another way to regulate the reactivity is the controlled release of initiator. This technique allowed Shi and coworkers (Hu et al., 2003) to obtain polyethylene grafted with 3-isopropenyl- α ; α -dimethylbenzene isocyanate (PE-graft-TMI) with fewer changes in molar mass and higher TMI content than the traditional route. This route did not modify monomer reactivity but caused changes in the reaction equilibrium because of the change in initiator concentration. Kamath and Pallys (Kamath and Pallys, 1990) reported in 1990 that using clays and other inorganic salts may reduce the degradation of polymers. They studied different peroxides and dispersion media to accomplish grafting of MA onto PP and PE. The reason, according to the authors, is the improved dispersion of peroxide in the polymer melt. Dispersing the peroxide in the medium reduces the local concentration of initiator, but at the same time improves the contact between peroxide and monomer, thereby increasing the reaction rate.

Shi and coworkers (Shi et al., 2006a, 2006b) conceived a method that allows controlling the initiator release and at the same time improves the dispersion of peroxide in the melt and adsorbs MA, increasing the reaction rate for the monomer (Shi et al., 2006). The authors reported an increase in grafting level, by means of titration, and decrease in molar mass reduction, by means of size exclusion chromatography (SEC). Muñoz and Bettini (Muñoz and Bettini, 2016) investigated this reaction using statistical methods and concluded that the grafting levels were not much affected by this route, however the authors used Fourier transformed infrared (FTIR) techniques for grafting level quantification. In addition, the reduction in molar mass was lower than that reported by Shi and coworkers (Shi et al., 2006a, 2006b). According to the authors, this was due to the presence of organically modified montmorillonite (o-MMT), which affects the recombination rate of formed macroradicals. Due to adsorption of chemicals, the macroradical concentration on o-MMT surface is high and improves the recombination of all kinds of radicals presents in the medium (Muñoz and Bettini, 2016, 2013).

It is clear there are many mechanisms involved in the system that need to be understood. Therefore, understanding and

optimizing this route may allow better control of polymer degradation during reactive processing. Therefore, we proposed the study of different ratios of organoclay:peroxide with the aim to understand if the amount of MMT affects the equilibrium of the reactions. The ratios studied were 0, 1:2, 1:1, 2:1 and 5:1. The other variables were maintained constant and the reaction products were analyzed by FTIR, SEC and parallel plate rheology.

2. Experimental protocols

2.1. Introduction of DCP in o-MMT

To insert peroxide into the clay, first 4 g of organically modified montmorillonite, Cloisite 20A (Southern Clays, USA), was dispersed under stirring in acetone p.a. (Synth, Brazil) with 5% of deionized water for 10 min. Then, an equivalent mass of dicumyl peroxide (DCP) (Perkadox BFCC, Akzonobel) diluted in pure acetone was added. The system was kept under stirring until dry (2–3 h). To remove excess of solvent and avoid excessive peroxide thermodegradation, samples were dried in a vacuum oven for 24 h at 40 °C. Four o-MMT:DCP mass ratios were produced: 1:2, 1:1, 2:1 and 5:1. The materials were characterized by X-ray diffraction to measure changes in o-MMT inter-gallery distance.

2.2. PP-graft-MA reactive processing

Polypropylene grafted with maleic anhydride (PP-graft-MA) was produced using polypropylene homopolymer HP502H (Quattor, Brazil) with a SEC measured $M_w = 905,000$ Da and $PI = 5.7$. As initiator, an amount equivalent to 1 phr of DCP was used pure and recrystallized with o-MMT (nanoreactors). For grafting, 2 phr of maleic anhydride p.a. (Sigma Aldrich, USA) was used. Reactive processing was carried out in a 60 mL Rheomix 600 (ThermoScientific) mixing chamber equipped with Roller 3600 rotors, at 180 °C, 50 rpm and N_2 flow. Firstly, 35 g of PP was melted in the mixing chamber (3 min), then DCP and MA were added simultaneously. The reaction time was set to 5 min, after which the chamber was opened, and the collected material quickly pressed and cooled to inhibit further reactions. Additionally, PP + o-MMT samples were prepared using the same protocol, but in absence of DCP and MA.

2.3. Purification of PP-graft-MA

Because of the presence of non-grafted maleic anhydride, all PP-graft-MA samples were purified. Next, 4 g of sample was dissolved in 150 mL of xylene p.a. (Synth, Brazil), using a reflux system at the solution boiling point (~ 130 °C) for 1 h. Then PP-graft-MA was precipitated by adding pure acetone. The precipitate was washed with acetone and dried in a vacuum oven for 24 h at 80 °C.

2.4. Titration

First, 1 g of purified sample was solubilized in 100 mL of xylene p.a. (Synth, Brazil) and cooled to 80 °C. The solution was titrated using phenolphthalein as indicator and KOH in isopropanol solution as titrant. An excess of 0.2 mL of KOH was added and a back-titration was performed using a 0.05 M HCl ethanolic solution. The total amount of grafted succinic groups ($\%m_{gt}$) was estimated using Eq. (1), where V represent the titrant volume and m_s is the sample mass:

$$\%m_{gt} = 9.8 \frac{[(V_{KOH} + 0.2)C_{KOH} - V_{HCl}C_{HCl}]}{m_s} \quad (1)$$

2.5. Calibration curve

A calibration curve for the vibrational mode at 1792 cm^{-1} was performed as described in the literature (Bettini and Agnelli, 2000). First PP was melt-mixed with a dodec-1-enyl succinic anhydride (Sigma-Aldrich, USA) in the Rheomix 600 mixer, at $180\text{ }^{\circ}\text{C}$, 100 rpm and N_2 flow. After 35 g of PP was melted for 3 min, then dodec-1-enyl succinic anhydride was added in the mixer chamber and mixed for an additional 2 min. The mixtures were hot-pressed for 2 min at $180\text{ }^{\circ}\text{C}$ and 6 ton of pressure, between $100\text{ }\mu\text{m}$ Kapton® films. These films were analyzed by FTIR and titration to determine the real concentration, some dodec-1-enyl succinic anhydride will be lost due to the high temperatures used.

The long succinic group calibration curve was estimated by the difference between the total succinic groups ($\%m_{\text{gt}}$), measured by titration, and the short grafted succinic groups ($\%m_{\text{g}}^{1792}$) estimated by FTIR and calibration curve with dodec-1-enyl succinic anhydride, as follows:

$$\%m_{\text{g}}^{1775} = \%m_{\text{gt}} - \%m_{\text{g}}^{1792} \quad (2)$$

where $\%m_{\text{g}}^{1775}$ is the mass of succinic groups with vibrational mode at 1775 cm^{-1} , $\%m_{\text{gt}}$ is the total succinic group content and $\%m_{\text{g}}^{1792}$ is the amount of grafted succinic groups with vibrational mode at 1792 cm^{-1} .

2.6. Homopolymerization of maleic anhydride

Two different MA homopolymers were prepared using protocols described by Trivedi and Curberlston (Trivedi, 1982). For the low molar product, 14 g of MA was dissolved in a flask containing 100 mL oxygen-free o-dichlorobenzene (Sigma-Aldrich) at $70\text{ }^{\circ}\text{C}$. The reaction was initiated by adding 0.18 g of DCP in 2 steps at an interval of 12 h (total reaction time = 48 h). The material was recrystallized using tetrahydrofuran (Synth, Brazil), dried and stored in a desiccator. The high molar mass material was synthesized by melting 30 g of MA and adding 1.8 g of DCP in six steps at 30-min intervals. Finally, the material was cooled and placed in a desiccator (total reaction time = 3 h).

2.7. FTIR characterization

The FTIR characterization was performed in a Nicolet 6700 FTIR spectrometer (ThermoScientific, UK), using 128 scan and resolution of 2 cm^{-1} . Polymeric film samples of $50\text{--}100\text{ }\mu\text{m}$ were produced by hot-pressing for 2 min at $180\text{ }^{\circ}\text{C}$ and 6 ton of pressure, between $100\text{ }\mu\text{m}$ Kapton films. The samples were dried for 24 h at $130\text{ }^{\circ}\text{C}$ to convert acid groups into anhydrides. For MA and synthesized oligomeric MA, samples were prepared as 0.1 wt% MA/KBr pressed pellet and analyzed using the same condition at which the polymeric films were analyzed. Prior to KBr pellet preparation, KBr (Pike, USA) was dried in a vacuum oven for 24 h at $150\text{ }^{\circ}\text{C}$.

2.8. Rheological characterization

To determine η_0 , all samples were analyzed by steady state rheology using AGR2 rheometer (TA Instruments, USA), equipped with a 25 mm diameter parallel plate. The samples were analyzed by varying the shear rate from 0.01 s^{-1} to 100 s^{-1} at $180\text{ }^{\circ}\text{C}$ under N_2 atmosphere.

2.9. Size exclusion chromatograph

To estimate PP-graft-MA molar mass, all samples were analyzed by size exclusion chromatography in a GPC-HT 350 chromatograph (Malvern, USA), equipped with refractive index detector and a kit

of guard columns + 3 mixed bed columns model HT-806 M (Shodex, Japan), operating at $140\text{ }^{\circ}\text{C}$. As solvent, chromatographic grade 1,2,4-trichlorobenzene (Tedia, USA) stabilized with 0.1 wt% of 2,6-bis(1,1-dimethylethyl)-4-methylphenol (Sigma-Aldrich, USA) was used. The main/auxiliary pump flow ratio was 1:2, applying 1 mL min^{-1} of solvent at the main pump. The equipment was calibrated using mixed PS standards from 1 kDa to 4 MDa and an $R^2 = 0.9997$ was obtained. The number of theoretical plates reported was about 30,000.

2.10. Scanning electron microscopy

For scanning electron microscopy (SEM) a Philips XL30 SEM FEG microscope equipped with energy-dispersive X-ray analyzer was used. The samples were cryogenically fractured and sputtered with gold.

3. Results and discussion

XRD was employed to verify the organically modified montmorillonite clay (o-MMT) intercalation by dicumyl peroxide (DCP) in all the samples, as shown in Fig. 1. As can be observed, neat o-MMT presents a diffraction peak at $2\theta = 3.3$, reported as the inter-gallery distance (Shi et al., 2006). When o-MMT is recrystallized in the presence of DCP, using different o-MMT:DCP ratios, this peak shifts to approximately 2.4° for all samples, indicating an increase in inter-gallery spacing. The increment in inter-gallery spacing is caused by the incorporation of DCP into the o-MMT structure, as reported by Shi and coworkers (Shi et al., 2006). Increasing the peroxide content results in the presence of three new peaks in the XR diffractograms (15.3° , 17.5° and 22.4°) related to DCP. For the sample with o-MMT:DCP ratio of 5:1 the DCP peak is not present in the XR diffractogram which suggests all DCP is located inside the o-MMT structure. Samples with o-MMT:DCP ratios lower than 5:1 present part of DCP adsorbed thus they act

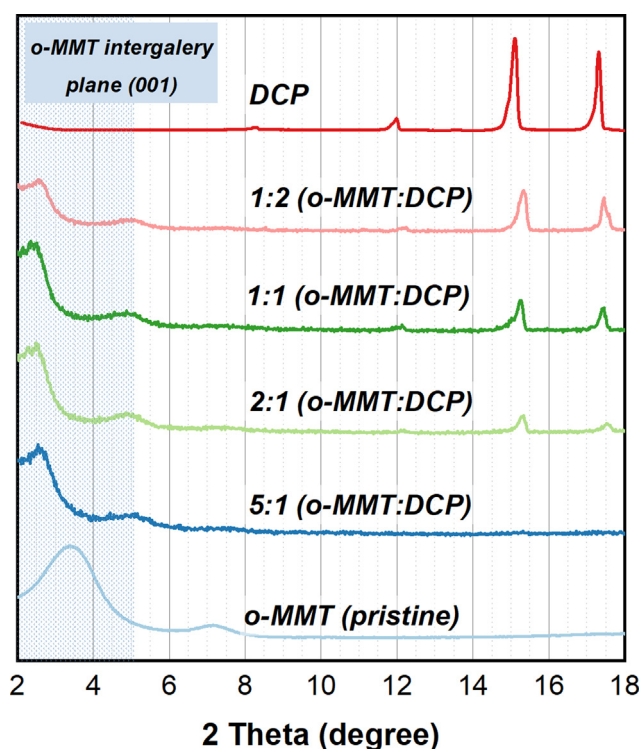


Fig. 1. XR diffractograms for o-MMT, DCP and their mixtures at different ratios.

more like a supporting structure than a nanoreactor, as previously proposed in the literature.

Fig. 1 reveals a change in the first o-MMT peak, from $2\theta = 3.4^\circ$ to $\sim 2.7^\circ$, for all o-MMT:DCP samples. This peak is attributed in the literature to the intergallery space of o-MMT (Shi et al., 2006a, 2006b) and the decrease in 2θ indicates an increase in the intergallery spacing. According to Bragg's law the increment in intergallery spacing, after o-MMT recrystallization in the presence of DCP, is about 1.05 nm, regardless of the o-MMT:DCP ratio used, indicating clay saturation. Since DCP diameter is about 0.935 nm, according to ACD/Labs software® simulation, this space allows the introduction of DCP with thickness of one DCP molecular diameter, i.e., no XRD pattern can be observed at complete absorption of DCP by o-MMT. When the intensity of peaks at $2\theta = 17.5^\circ$ is plotted as a function of DCP mass concentration, a quasi-linear correlation is observed (Fig. 2). According to the fitted linear function, o-MMT can absorb up to 16 wt% of DCP, thus the o-MMT reported in the literature probably did not take advantages of diffusional peroxide release as proposed. Samples with o-MMT:DCP ratio of 5:1, present 16.7 wt% of DCP, thus all DCP was considered to be absorbed by the clay structure.

After reactive processing, samples of PP-graft-MA produced were cryogenically fractured and analyzed by SEM-EDS. Fig. 3(a) and Fig. 3(b) show images from sample PP-graft-MA produced using o-MMT:DCP ratio of 5:1. Fig. 3(a) reveals the presence of a high number of particles across the fractured surface, showing a good distribution of o-MMT in PP-graft-MA matrix with poor dispersion. Muñoz and Bettini observed similar behavior in systems using PP/o-MMT and PP/o-MMT/DCP (Muñoz and Bettini, 2013), i.e., o-MMT presents good distribution and poor dispersion in PP matrix. Fig. 3 (b) shows an EDS analysis of one of o-MMT tactoids, evidencing the poor dispersion of o-MMT in PP-graft-MA, that is corroborating the hypothesis of diffusional control of the reaction for samples using o-MMT:DCP ratio of 5:1.

To verify the effect of using o-MMT during reactive processing all samples were characterized by FTIR, presented in Fig. 4. In all samples, the absorption band of the vibrational mode close to 1792 cm^{-1} is present, attributed to succinic groups grafted onto PP chains (De Roover et al., 1995). FTIR spectra clearly show the

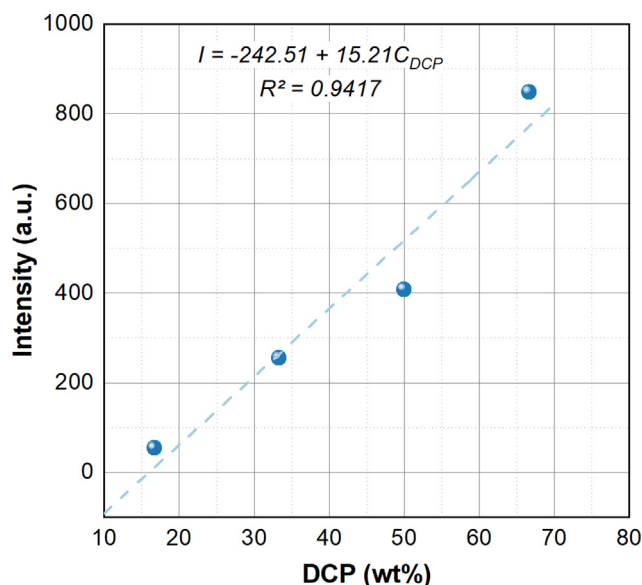
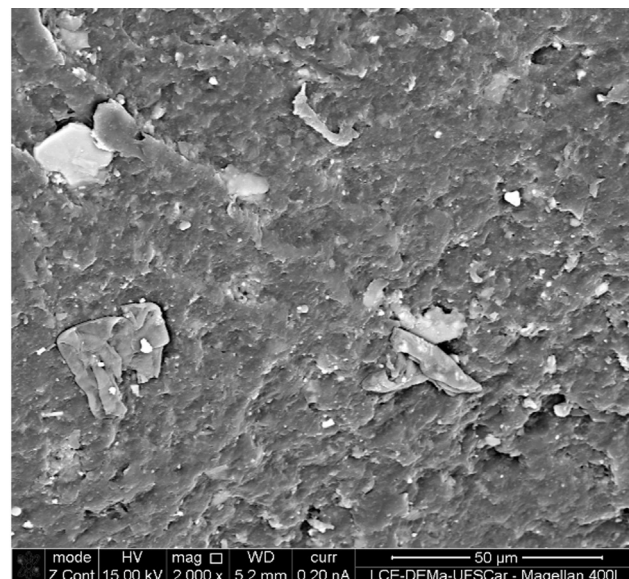
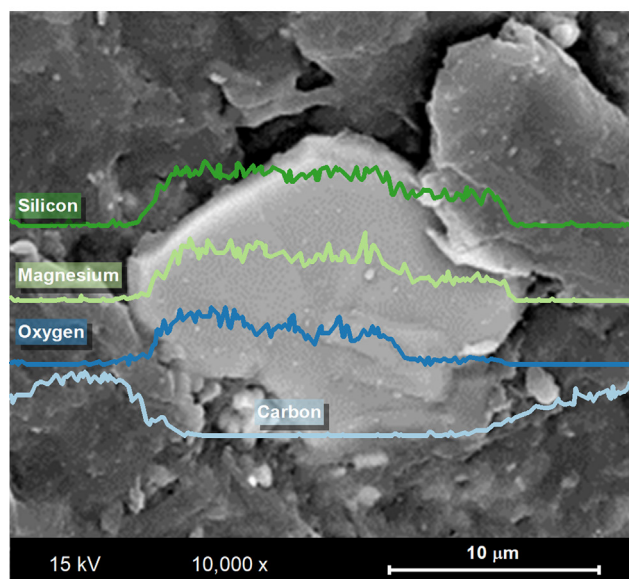


Fig. 2. Correlation between XRD signal and DCP mass concentration for the o-MMT/DCP nanoreactors.



(a)



(b)

Fig. 3. SEM images of PP-graft-MA produced using o-MMT:DCP ratio of 5:1. (a) 2,000x magnification. (b) 10,000x magnification and EDS line analysis of o-MMT tactoid.

existence of a band at 1775 cm^{-1} for all samples, and this vibrational mode is more evident for samples processed in the presence of o-MMT. Sclavons and coworkers (Sclavons et al., 2005) reported a vibrational mode at 1785 cm^{-1} for PP-graft-MA obtained by ene-reaction route, which was assigned to the interaction between the vicinal succinic groups formed by this route. Using grafting assisted by styrene, Bettini and Ruvolo-Filho (Bettini and Ruvolo Filho, 2008) reported the grafted succinic vibrational mode shifts to 1780 cm^{-1} and assigned this to the grafting of an alternated maleic anhydride-styrene copolymer. In this way it is possible to know which vibrational mode observed at 1775 cm^{-1} results from a strong interaction between succinic groups, including the grafting of long succinic chains. To explore this hypothesis two polymerization routes for maleic anhydride were studied:

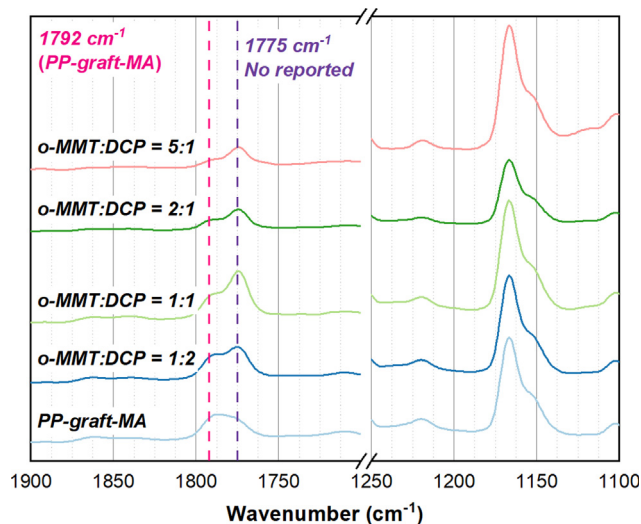


Fig. 4. FTIR spectra for PP-graft-MA produced via reactive processing using different o-MMT:DCP ratios and reference produced in absence of o-MMT (w/o o-MMT:DCP, light blue line). (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

- (1) **Solution polymerization:** aimed to produce low molar mass products.
- (2) **Bulk polymerization:** intended to produce higher molar mass products.

As shown in Fig. 5 (a)-(c), the solution-polymerized product presents a light brown color while the bulk-polymerized product is brown, in contrast with the white-colored MA. The FTIR spectra (Fig. 5 (d)) show significant differences for the two polymerized MA. In both cases the MA band, located at 1705 cm^{-1} and assigned to free maleic anhydride, was shifted to higher values according to the polymerization degree. Related to the bulk-polymerized product, the band is shifted to 1776 cm^{-1} , very close to the band observed for PP-graft-MA. The refractive index curve versus retention time from SEC of polymerized MA, presented in Fig. 5 (e), reveals that both materials have molar mass lower than 1,000 Da. Because molar masses are too low, it was not possible to calculate the molar mass of the materials, but a significant difference can be observed in retention time curve regarding peak and width, indicating a high molar mass of the bulk polymerized MA. Thus, the band observed at 1775 cm^{-1} for PP-graft-MA is probably caused by maleic anhydride oligomers grafted onto PP backbone.

The number of succinic groups grafted onto PP was estimated using FTIR and classical volumetric analysis, to obtain the calibration curves presented in Fig. 6 (a) and Fig. 6 (b). The extension of PP chain degradation was assessed using parallel plate rheology and SEC. Fig. 6 (c) shows the effect of o-MMT in the amount and type of grafted succinic groups and the changes in shear zero viscosity (η_0) for the produced PP-graft-MA. As can be observed the number of grafted succinic groups tend to reduce with the increase in o-MMT content, but all routes lead to a predominance of long succinic grafts (around 70–75% of long succinic groups). Since grafting is not the preferential reaction, an increase in PP β -scission reaction was expected, however, as can be observed in Fig. 6 (c) the

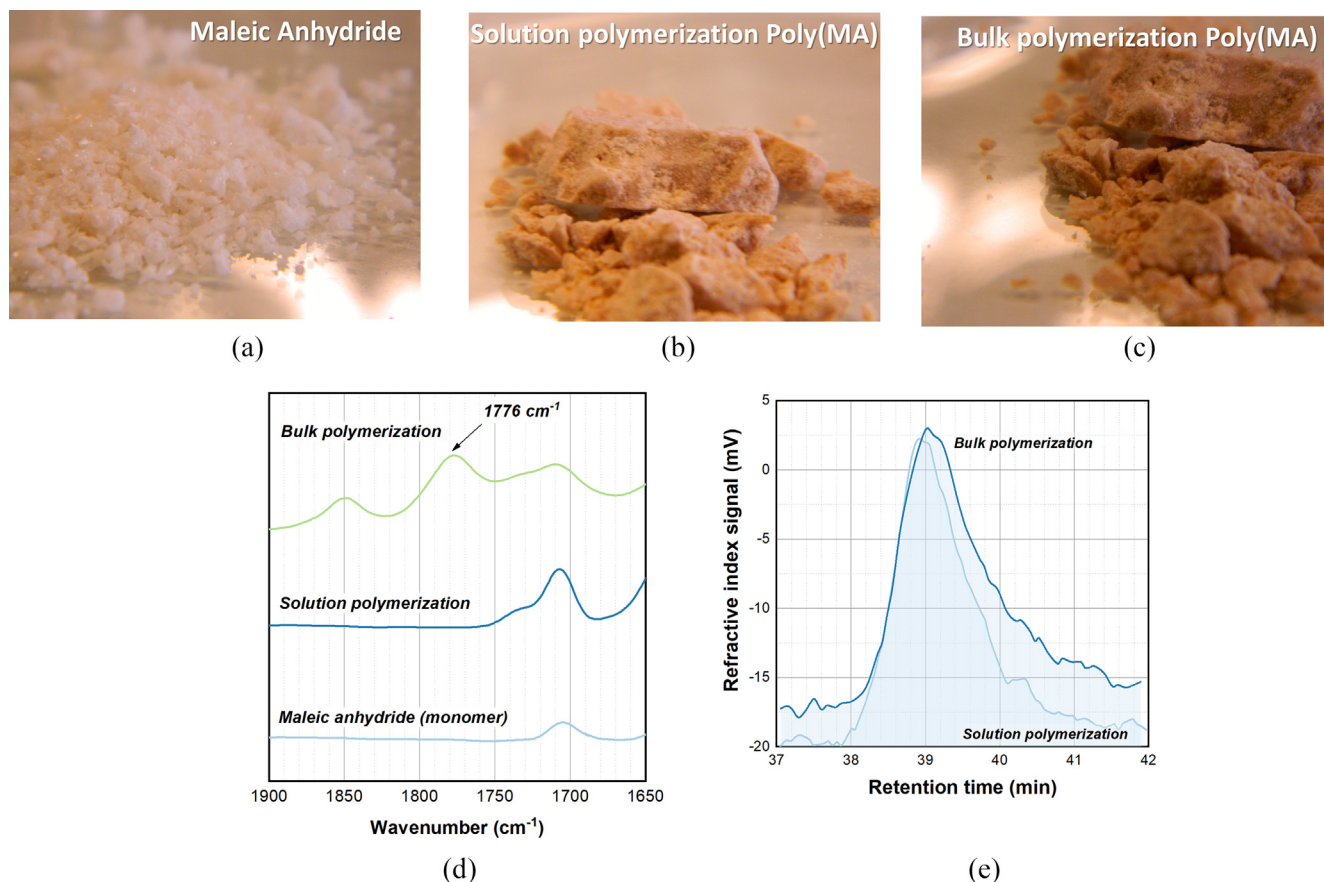


Fig. 5. Poly(maleic anhydride) characterization: (a)-(c) Pictures of MA and poly(MA); (d) FTIR spectra of maleic anhydride and poly(MA) (e) SEC curve for poly(MA).

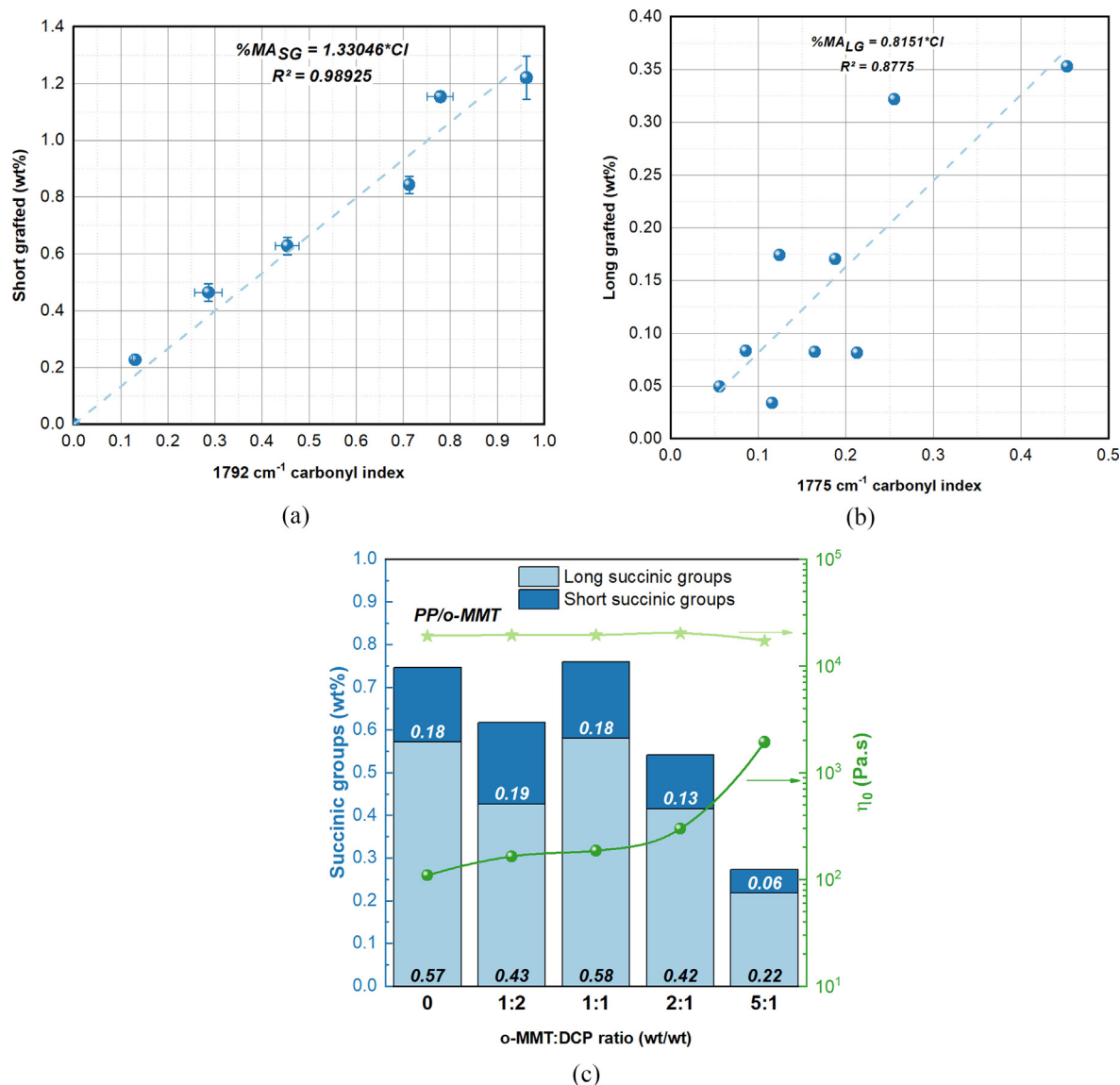


Fig. 6. Calibration curves for (a) short (IC^{1792}) and (b) long (IC^{1775}) succinic groups of PP-graft-MA. (c) Mass percentage of grafted maleic anhydride and zero shear-rate viscosity for PP-graft-MA; the numbers indicate the amount of short or long grafted succinic groups.

measured η_0 increases with increasing o-MMT content. As shown in the curve denoted as PP/o-MMT in Fig. 6 (c), the effect of o-MMT on η_0 can be neglected, thus the observed changes in η_0 for the PP-graft-MA are probably related to changes in the reaction. The molar mass data, summarized in Table 1 corroborates the rheological measurements, indicating reduced PP chain degradation in the presence of o-MMT. In Fig. 7 the theoretical Mark-Houwink-Sakurada relation for PP is shown. The distribution of PP-graft-MA data along theoretical curve indicates the materials are quite

linear, presenting short branches and low effect of succinic group interactions, since lower than 0.66 wt% of MA was grafted onto PP.

The fact that PP-graft-MA chains are linear implies that the grafting reaction occurs preferentially at secondary carbons of the PP chain, and not at the tertiary ones as proposed in the literature (Shi et al., 2006a, 2006b). Despite the fact that the molar mass reduction decreases in the presence of o-MMT, the minimum reduction in M_w observed was higher than 55%, indicating PP β -scission is still the preferential reaction. Considering all DCP was

Table 1

Molar mass and rheological data of PP-graft-MA and neat PP.

o-MMT:DCP ratio (wt/wt)	η_0 (Pa.s)	M_n (g/mol)	M_w (g/mol)	PD
HP502H (Reference)	20,790	159,000	905,000	5.7
0:1	109	96,900	188,000	1.9
1:2	164	94,300	193,000	2.0
1:1	185	55,600	164,000	3.3
2:1	298	103,000	303,000	2.9
5:1	1,930	153,000	402,000	2.6

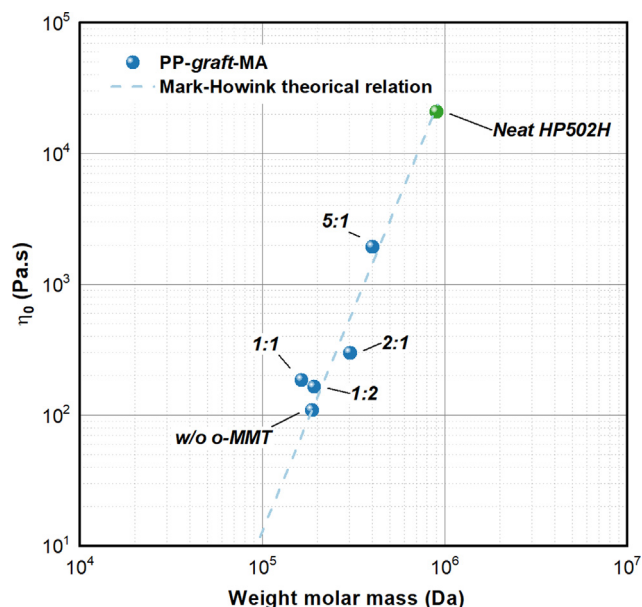
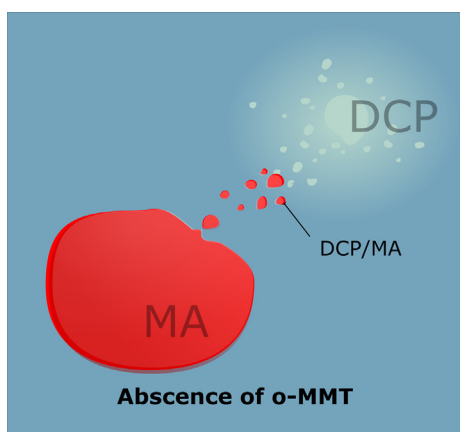
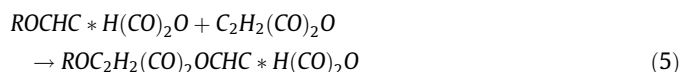
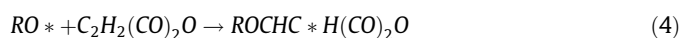


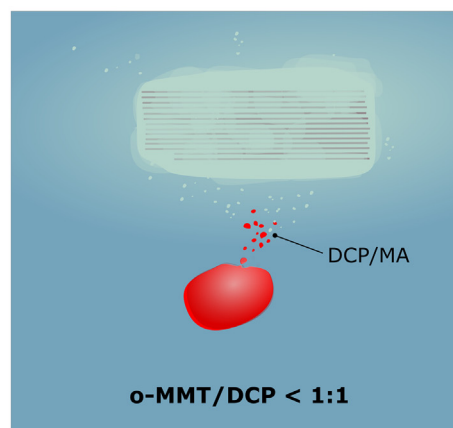
Fig. 7. Mark-Howink-Sakurada relation for PP and PP-graft-MA.

decomposed, the lower reduction in molar mass can be attributed to a higher number of PP macroradical recombination, as previously proposed (Muñoz and Bettini, 2016, 2013), and MA homopolymerization.

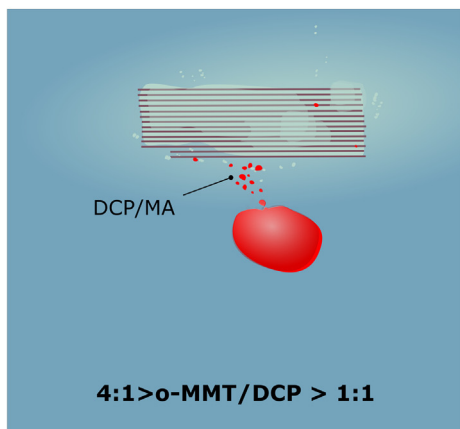
Fig. 8 shows a proposed mechanism for o-MMT in this system. When MA and DCP are mixed with PP, they tend to form second phases due to lack of polarity (Moad, 1999). During melt mixing mass transfer processes occur between DCP and MA phases (Fig. 8 (a)), because of their similar polarities and, DCP undergoes thermal degradation (Eq. (3)). As a result of DCP and MA compatibility, oligo (MA) and MA radicals are formed at the DCP/MA interface, as shown in Eqs. (4) and (5). At the same time alkoxy radicals abstract tertiary H from PP, as shown in Eq. (4). If tertiary PP macroradicals do not react, PP undergoes β -scission (Eq. (6)). All radicals formed react with each other according to the reactions shown in Eqs. (7)–(13).



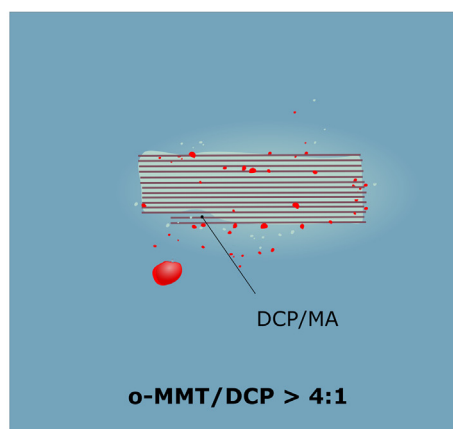
(a)



(b)

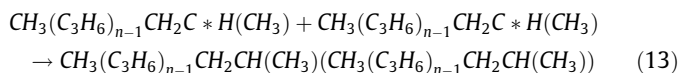
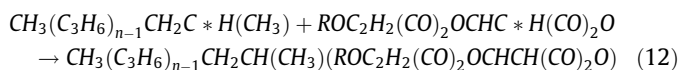
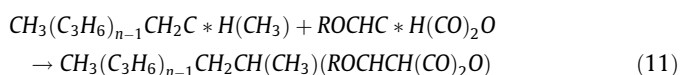
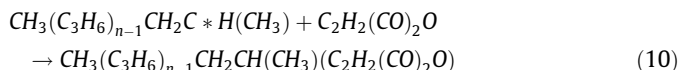
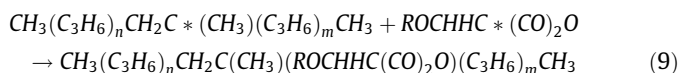
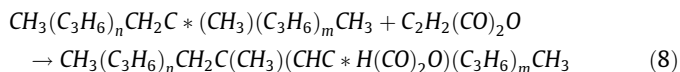
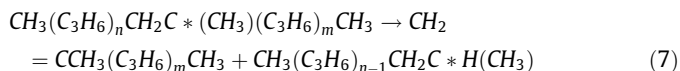
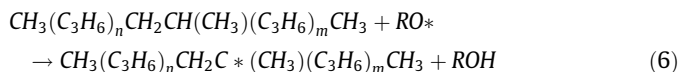


(c)



(d)

Fig. 8. Proposed mechanism of the role of o-MMT when used as DCP support at different o-MMT/DCP ratios.



The introduction of long succinic chains at tertiary PP macroradicals probably does not occur due to the high level of hindrance imposed by the presence of the oligo(MA), so only its reactions with secondary PP macroradicals were considered. As DCP is not well dispersed in PP, because of polarity, DCP drops will be surrounded by regions rich in PP macroradicals. Considering there is no MA in these regions, β -scission (Eq. (7)) takes place followed by secondary PP macroradical recombination, as shown in Eq. (13).

The presence of o-MMT has different effects on the reactional system, depending on the proportions used. Fig. 8 shows how different o-MMT/DCP ratios can change the o-MMT role during reactive processing. Fig. 8(a) shows the interphase in the absence of o-MMT. In this situation DCP distribution depends on its partition coefficient (Moad, 1999). Reactions occur according to those previously described. Fig. 8 (b) reveals the interface generated at o-MMT:DCP ratios < 1:1. In this case, all o-MMT is covered by DCP, which hampers its interaction with MA, thereby reducing oligo (MA) and MA radical formation. At the same time, PP/DCP interphase increases which leads to PP β -scission followed by secondary PP macroradical recombination at the clay surface, as proposed by Muñoz and Bettini (Muñoz and Bettini, 2013).

When o-MMT:DCP ratio is increased (Fig. 8(c)), part of o-MMT is not covered by DCP, allowing its interaction with MA. The probability of o-MMT/MA interaction increases the reaction between MA and DCP, increasing oligo (MA) formation. As a result, the proportion of long succinic chain grafts increases as shown. As MA oligomerization consumes the alkoxy radical, a reduction in PP chain scission is observed.

In a situation where all DCP is inside the o-MMT structure, the whole system will be controlled by diffusional release of alkoxy radical and the o-MMT surface will be free to interact with MA. The o-MMT/MA interaction results in an increase in oligo (MA), because o-MMT attracts MA to a surface covered with alkoxy radicals. The extension of MA oligomerization increases molecule hindrance and consumes the alkoxy radicals, reducing grafting level, due to high hindrance and low PP macroradical concentration at o-MMT surfaces. Because of this grafting level, PP degradation tends to drop when high amounts of o-MMT are used.

4. Conclusion

The use of o-MMT for controlled peroxide release during reactive processing is possible only when using large amounts of clay. For DCP this limit is around 16 wt% of peroxide, i.e., o-MMT concentration higher than 86 wt%. According to physicochemical characterizations, for this system large amounts of grafted material present long succinic chains grafted onto PP backbone through secondary PP macroradicals, generating linear products. The use of large amounts of o-MMT leads to massive production of oligo (MA) that consumes the initiator. Because the alkoxy radicals are consumed by MA oligomerization, PP degradation reduces, and due to increased hindrance imposed by oligo (MA) chains, the grafting level will also drop. Therefore, the use of o-MMT acts as a heterogeneous catalyst, as it adsorbs MA, promoting its polymerization and reducing PP degradation and grafting.

CRediT authorship contribution statement

Pablo Andrés Riveros Muñoz: Conceptualization, Methodology, Formal analysis, Investigation, Writing - original draft, Writing - review & editing.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Data availability statement

The raw/processed data required to reproduce these findings cannot be shared at this time due to technical and time limitations.

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